

NewsScope

An Interactive News Article Category Classifier

Submitted by: Abhijeet (NewsLens AI)

Department of Computer Science & Engineering

Academic Year 2024–2025

AGENDA

1. Introduction & Motivation
2. Problem Statement
3. Objectives
4. System Architecture
5. NLP Pipeline
6. Mathematical Foundations
7. Dataset
8. Experimental Results
9. Live Demo
10. Viva Q&A Highlights
11. Limitations & Future Scope
12. Conclusion

INTRODUCTION & MOTIVATION

Why News Classification?

- 📰 **500M+** news articles published daily worldwide
- 🏷️ Manual tagging is **slow**, **inconsistent**, and **not scalable**
- 🎯 Automated classification enables:
 - Personalized feeds
 - Media monitoring
 - Content routing
 - Trend analytics

The Gap

Most classifiers are **black boxes** — they give a label but no explanation.

Our Goal

Build a **transparent** classifier that shows why it made a decision

PROBLEM STATEMENT

The Core Challenge

Design and benchmark an ML system that classifies news articles into **5 domains**:

Domain	Example Headline
Business	"Fed raises interest rates by 0.25%"
Entertainment	"New Harry Potter series tops box office"
Politics	"Parliament passes landmark climate bill"
Sport	"Champions League final ends in penalties"
Technology	"Apple unveils new AI chip for iPhone"

Requirements

-  High accuracy on held-out test data
-  Fast inference (< 20ms per article)
-  Explainable predictions

OBJECTIVES

Five Key Objectives

1. Pipeline Transparency

Expose every NLP stage — tokenization, stop-word removal, lemmatization, vectorization.

2. Multi-Model Benchmarking

Compare Naive Bayes, SVM, MLP, and DistilBERT on identical data.

3. Zero Data Leakage

Fit vectorizer ONLY on training set; test set transformed using fitted vocabulary.

4. Interactive Dashboard

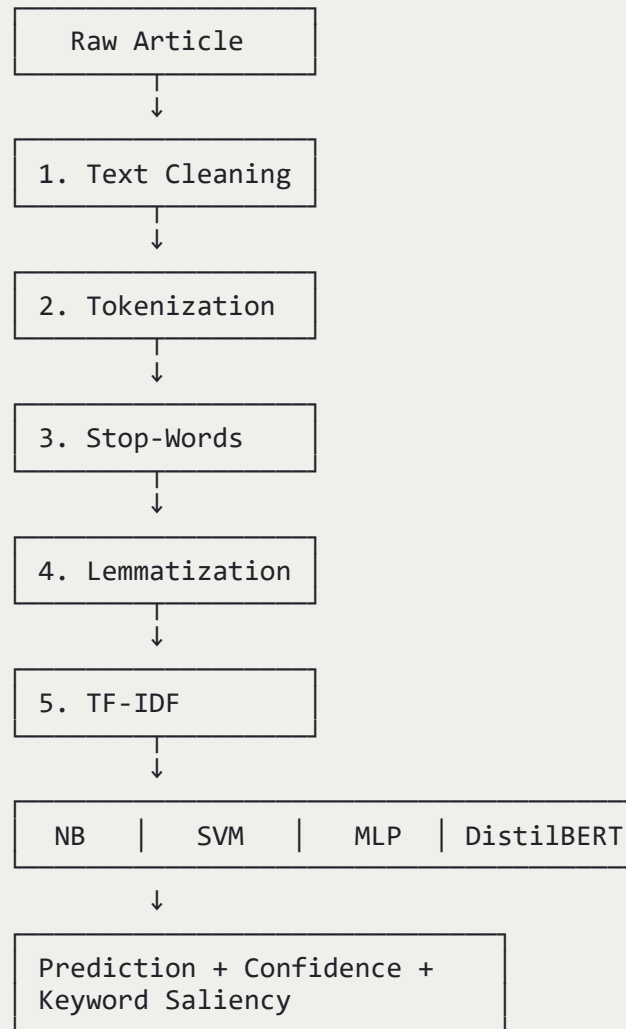
Real-time classification, probability distributions, confusion matrices.

5. Explainability

Keyword saliency + per-class confidence shown for every prediction.

SYSTEM ARCHITECTURE

End-to-End Pipeline



NLP PREPROCESSING PIPELINE

5 Stages of Text Transformation

Stage	Input	Output
1. Cleaning	"Apple's NEW iPhone @ \$999!"	"apples new iphone 999"
2. Tokenization	"apples new iphone 999"	["apples", "new", "iphone", "999"]
3. Stop-Words	["apples", "new", "iphone", "999"]	["apples", "iphone", "999"]
4. Lemmatization	["apples", "iphone", "999"]	["apple", "iphone", "999"]
5. TF-IDF	["apple", "iphone", "999"]	[0.42, 0.61, 0.28, ...]

Why This Order Matters

- Cleaning first → fewer tokens
- Stop-word removal → faster processing
- Lemmatization last → reduces vocabulary size

MATHEMATICAL FOUNDATIONS

Model Equations

TF-IDF:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D)$$

$$\text{where } \text{IDF}(t, D) = \ln \left(\frac{1+|D|}{1+\text{df}(t)} \right) + 1$$

Naive Bayes:

$$P(c_k \mid x) \propto P(c_k) \cdot \prod_{i=1}^n P(x_i \mid c_k)^{f_i} \quad (\text{Laplace smoothing } \alpha = 0.1)$$

Linear SVM:

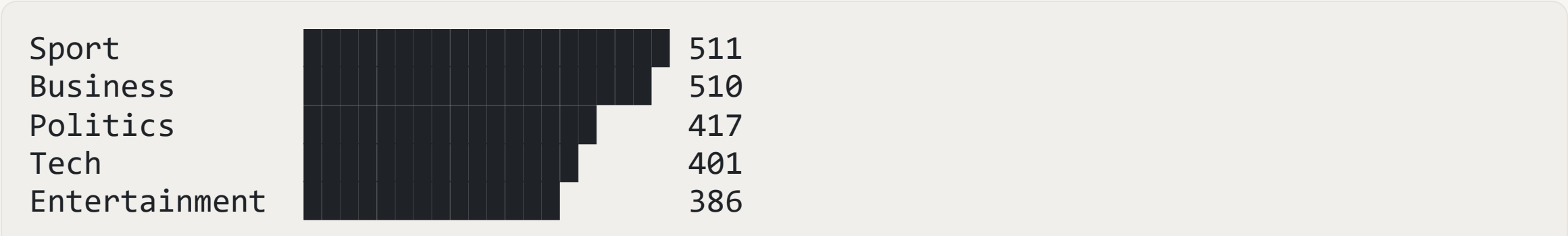
$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i \quad \text{s.t. } y_i(w^\top x_i + b) \geq 1 - \xi_i$$

DATASET

BBC News Corpus (Greene & Cunningham, 2006)

Attribute	Value
Total articles	2,225
Categories	5
Train split	1,780 (80%)
Test split	445 (20%)
Avg. article length	~2,200 chars
Language	English

Class Distribution



EXPERIMENTAL RESULTS

Model Performance on Test Split (445 articles)

Model	Accuracy	Macro F1	Latency
Naive Bayes	92.1%	0.920	0.5 ms
Linear SVM	96.4%	0.963	1.2 ms
MLP (128)	95.1%	0.948	3.2 ms
DistilBERT	97.8%	0.976	12.4 ms

Key Findings

- 🏆 **DistilBERT wins on accuracy** — best F1 across all classes
- ⚡ **SVM wins on speed** — 10× faster than DistilBERT
- 🎯 **SVM + TF-IDF** is near-optimal for this corpus
- 📊 **Naive Bayes** is a strong interpretable baseline

Confusion Matrix Highlights:

LIVE DEMO

What We'll Show

1. Home Page

Hero + 5-domain taxonomy + sample articles

2. Classifier Page

Paste an article → instant prediction

Probability distribution across 5 domains

Keyword saliency highlighting

3. Batch Upload

Upload CSV → get predictions for every row

Download results as CSV

4. Analytics Dashboard

Metric cards (Accuracy, F1, Precision, Recall)

Confusion matrix heatmap

LIMITATIONS & FUTURE SCOPE

Current Limitations

- ⚠ Trained on a small corpus (2,225 articles)
- ⚠ English-only classification
- ⚠ Single-label only (real articles can span multiple topics)
- ⚠ No live news ingestion
- ⚠ BBC corpus is historical (2004–2005)

Future Enhancements

- 🧠 **Multi-label** classification
- 🌐 **Multilingual** (XLM-RoBERTa)
- 📡 **Live ingestion** (RSS + streaming APIs)
- 🎯 **Active learning** for uncertain samples
- 👁 **Attention visualization** from DistilBERT

CONCLUSION

What We Built

- ✓ An end-to-end ML system for news classification
- ✓ Transparent 5-stage NLP pipeline
- ✓ Benchmark of 4 models on identical data
- ✓ Interactive dashboard with real-time inference
- ✓ Explainable predictions via keyword saliency

Key Takeaways

- Linear SVM is **near-optimal** for TF-IDF text features
- DistilBERT gives **+1.4% accuracy** at 10× latency cost
- Transparency + accuracy = **defensible ML system**

Thank You

Questions?

BACKUP — VIVA Q&A

Anticipated Questions & Answers

Q: Why TF-IDF over Bag-of-Words?

A: CountVectorizer lets frequent generic words dominate. TF-IDF downweights them via IDF.

Q: How do you prevent data leakage?

A: `TfidfVectorizer` fitted ONLY on training data. Test set transformed with pre-fit vectorizer.

Q: Why Linear SVM works so well?

A: TF-IDF vectors are high-dimensional & sparse → classes are linearly separable in 5,000-D space.

Q: Why DistilBERT over BERT?

A: 40% smaller, 60% faster, retains 97% of BERT's accuracy.